
King County Housing Data and Insights of customer services

by Joscha Metz

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The Client

I had a quick chat with the client to investigate following statements.

People

The client wants to have 3 bedrooms for both of his houses.

Central Location

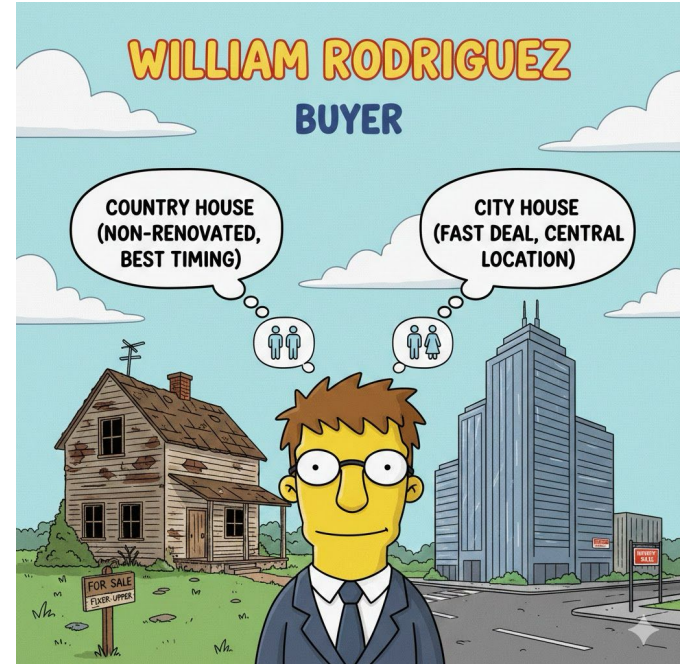
Following zip codes are required:

98109, 98119, 98101, 98104, 98101, 98154, 98102

98112, 98122, 98134, 98144

Prices

The client is not aware of the price differences of his searchings.



– What is the KCHD?

About the building

- qty of bedrooms
- qty of bathrooms
- ft² of the home
- ft² of the lot
- total floors
- ft² basement
- building year
- year house was renovated

the location

- zip code
- latitude coordinat
- longitude coordinate
- living space of nearest 15 neighbors
- lot space of nearest 15 neighbors

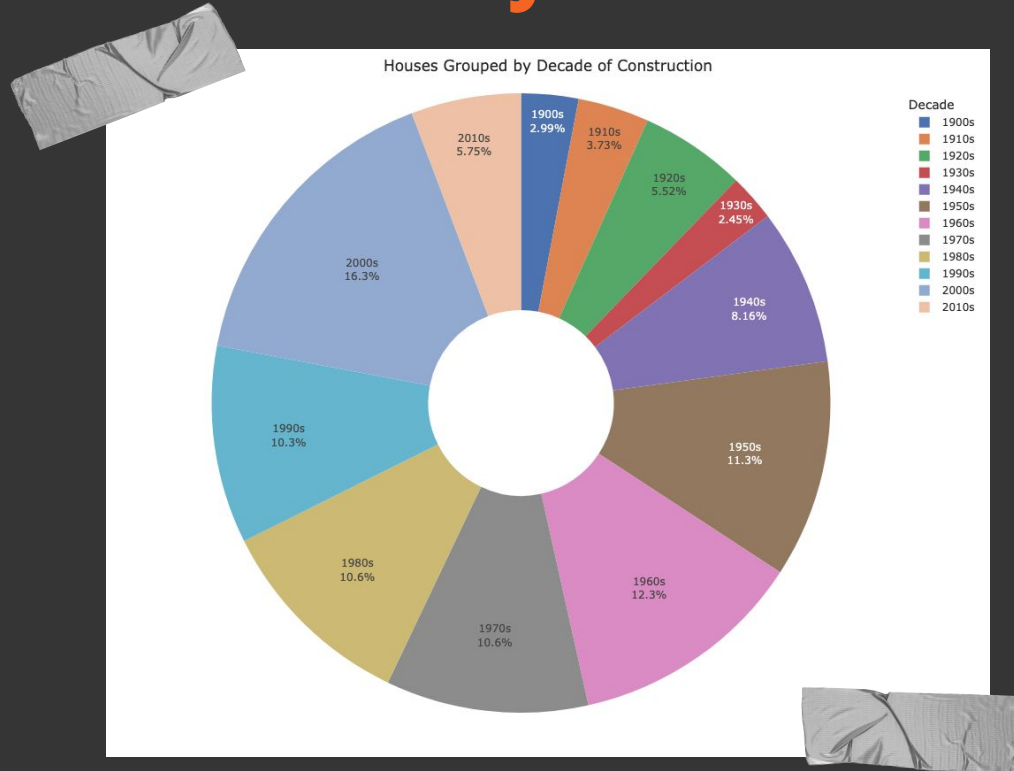
sales information / condition

- price
- date of sale
- view to waterfront
- quality of view
- overall condition (1-5)
- KC grading system

information about > 20k houses

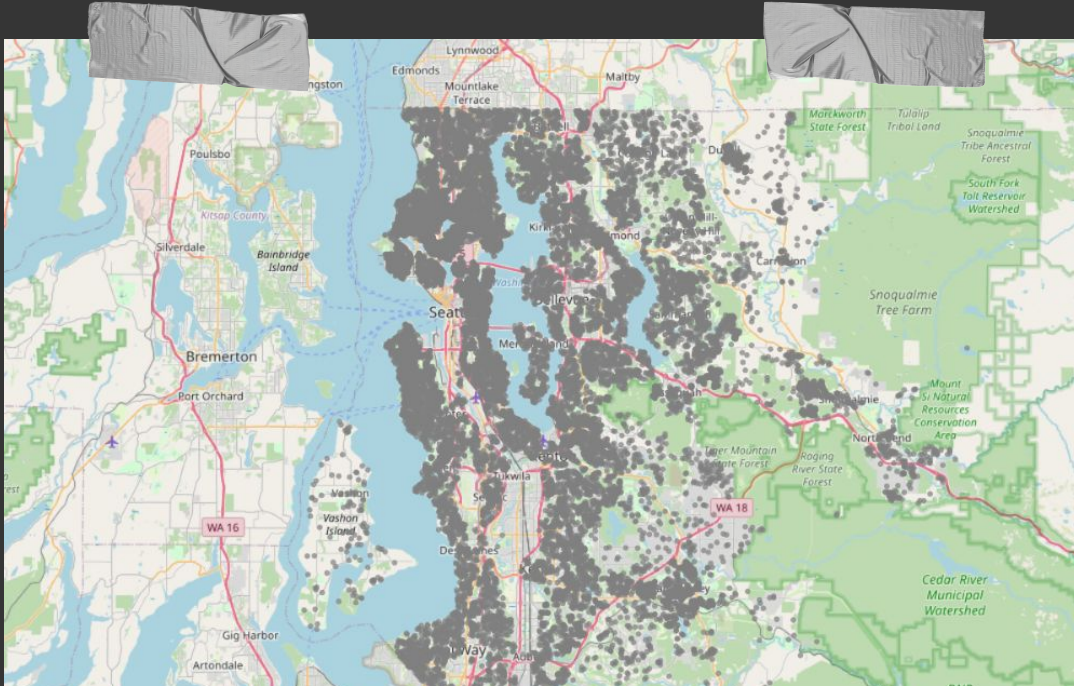
– What is the KCHD?

In which century the houses were built.



– What is the KCHD?

Where are the houses located?

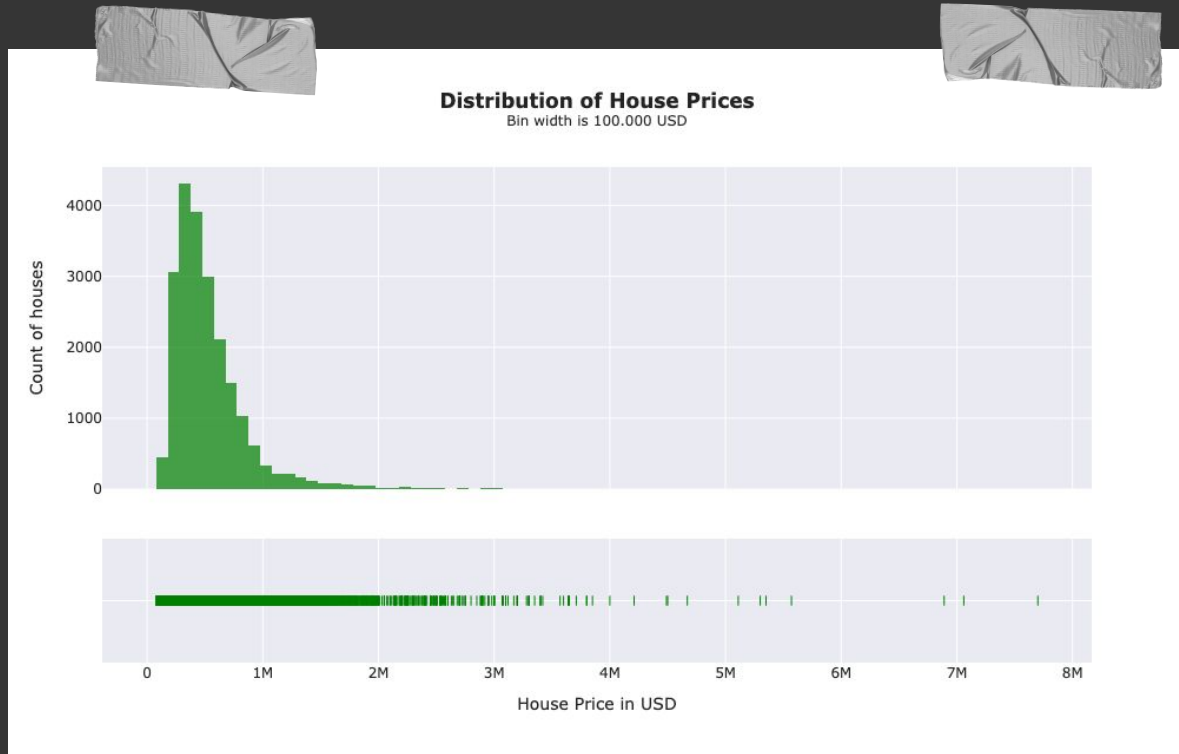


The Rural-Urban
Commuting Area
(RUCA) Codes were
incorporated to
categorize between
country and city houses.

– What is the KCHD?

The House Prices from 05/2014 to 05/2015

Expensive houses above 4 million USD are ignored in further pricing investigations.



– Hypothesis

Hypo 1:

There is a specific time of year when prices are significantly higher or lower.

Hypo 2:

The price of a country house does not increase significantly post-renovation.

Hypo 3:

A city house with 3 bedrooms is significantly more expensive than a similar house in a less central location.

- Hypothesis

Hypo 1:

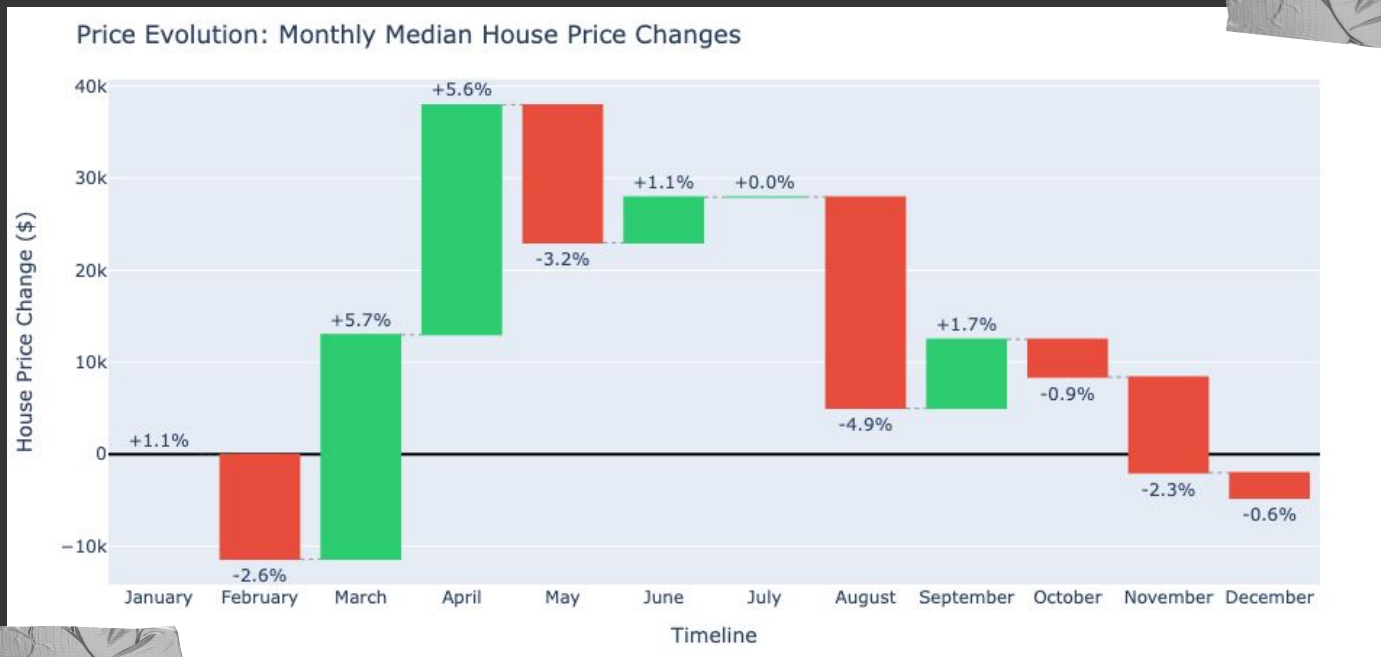
There is a specific time of year when prices are significantly higher or lower.

Where to look at?

- House Prices
- date of sale

How to look at?

- House Prices as median of the month
- add the change to the previous month
- months sorted by year



- Hypothesis

Hypo 2:

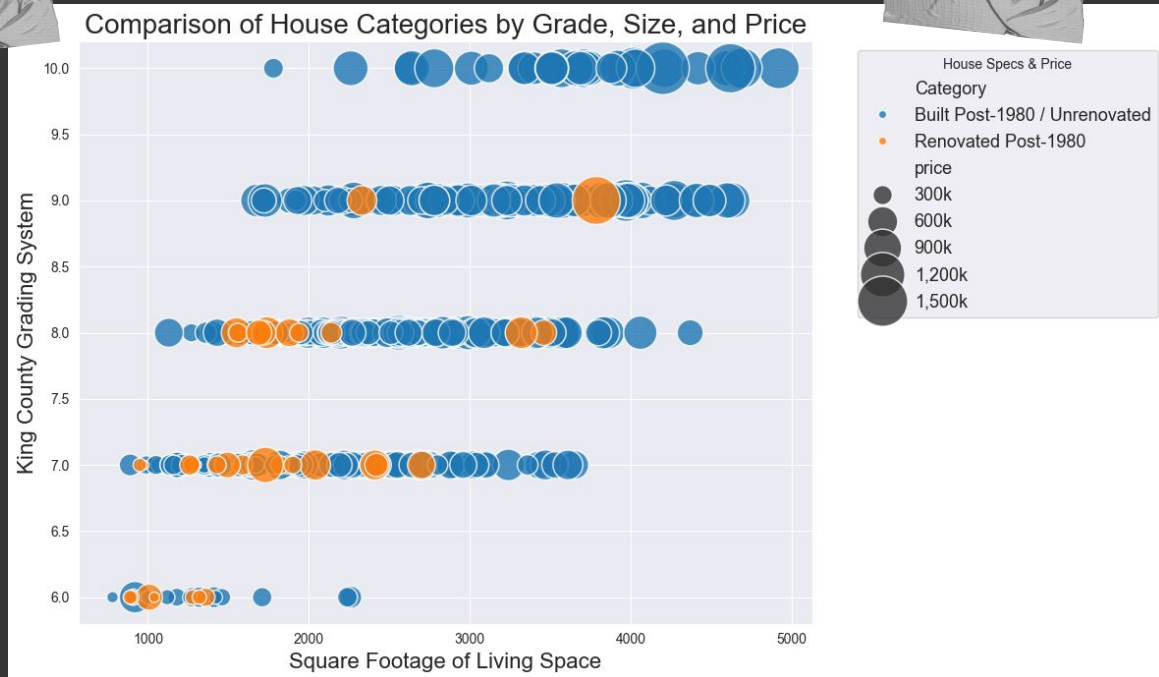
The price of a country house does not increase significantly post-renovation.

Where to look at?

- House Prices
- King County grading system
- ft² of the home
- year of built or renovation

How to look at?

- reduce dataset to
 - built/renovated > 1980
 - KC grading in between 5-10
 - ft² of the home < 5000
- adding RUCA
- prices shown in size of bubbles



- Hypothesis

Hypo 3:

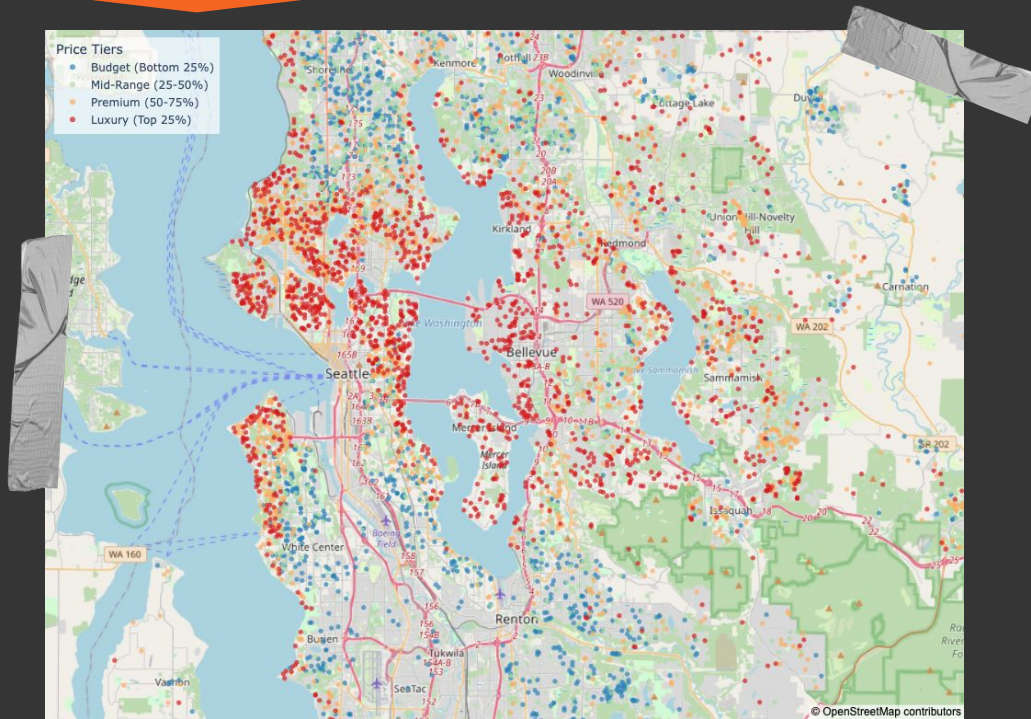
A city house with 3 bedrooms is significantly more expensive than a similar house in a less central location.

Where to look at?

- House Prices
- longitudinal and lateral coordinates

How to look at?

- getting the map from OpenStreetMap 🙏
- reducing the dataset to houses with 3 bedrooms



My Recommendations

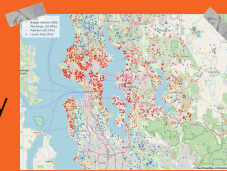
Time to Buy

Buy a city house now! Stay informed for a country house and buy it around January 2027.



How to split your invest

Country houses are much cheaper especially if not with lakeview. Split your invest in around 30% country house and 70% city house per ft².

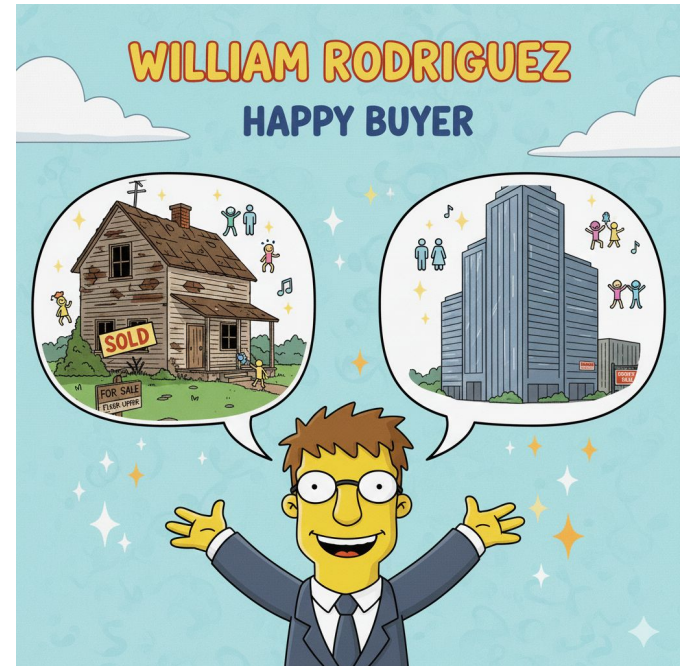


Renovated or not?

Have a deeper look at the King County House grading system and the condition of the house.

Defining an object

Meet up with my client again and look at specific houses.



Raise your hand and ask!

Thanks for listening.

Please have a deeper look in my repository: [ds-eda-project](#)

visit my github: [joschamz](#)



Tip for myself

Start your presentation early and dive deep when time is on your side.